

Department of Electrical & Electronic Engineering
Brac University
Semester: Spring 25



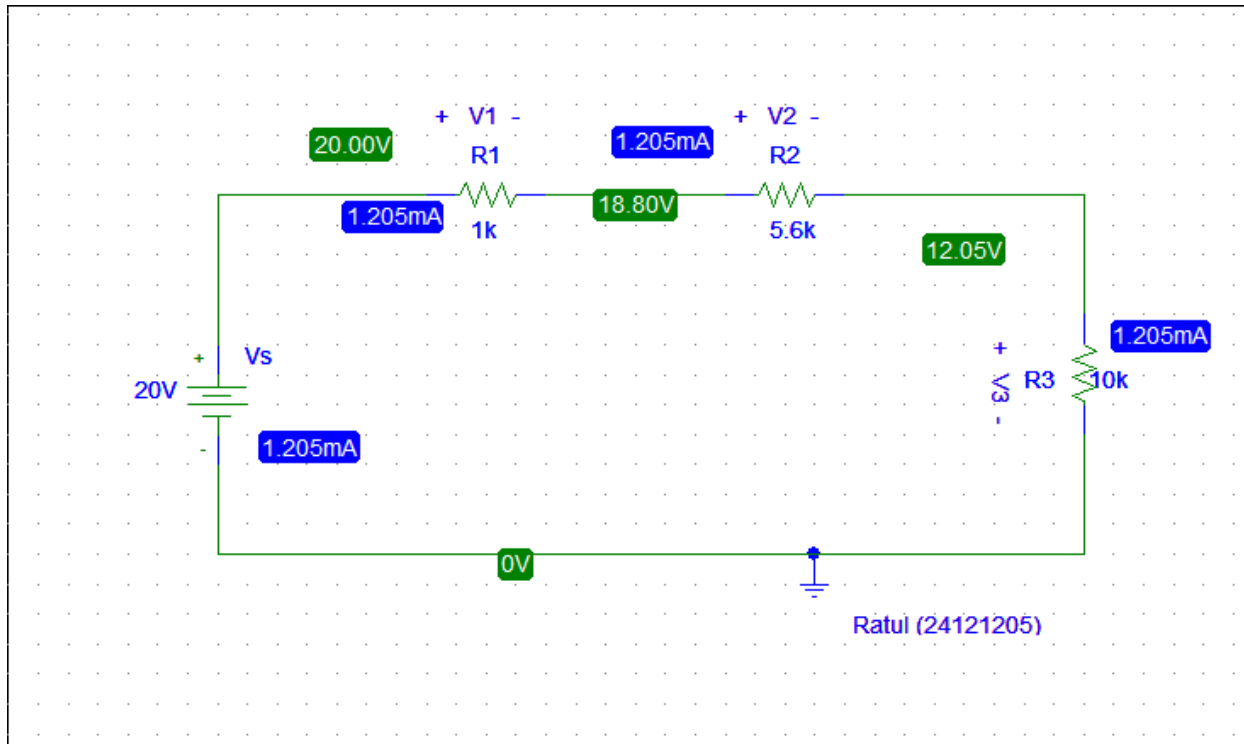
EEE101L
ELECTRICAL CIRCUITS I LABORATORY

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Section:05

Experiment No. 1

Verification of KVL and KCL

Part-A: Verification of KVL



Here,

$$V_s = 20V$$

$$V_1 = (20 - 18.80) = 1.2V$$

$$V_2 = (18.80 - 12.05) = 6.75V$$

$$V_3 = (12.05 - 0) = 12.05V$$

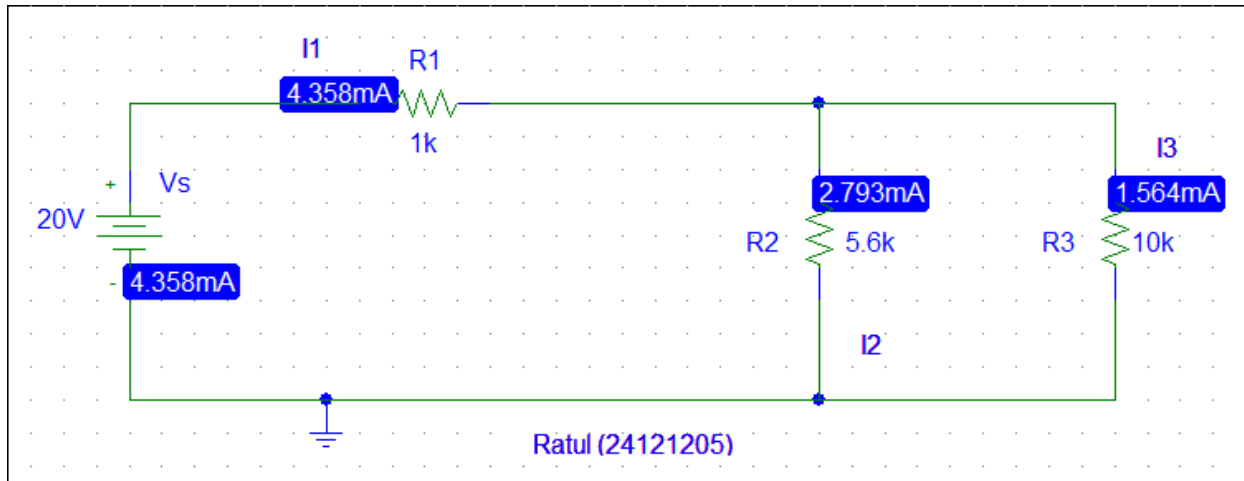
$$\text{Now, } V_s = V_1 + V_2 + V_3$$

$$= (1.2 + 6.75 + 12.05)V$$

$$= 20V$$

[KVL Proved]

Part-B: Verification of KCL



Here ,

I_1 - 4.358mA

I_2 - 2.793mA

I_3 - 1.564mA

Now, $I_1 = I_2 + I_3$

$$= (2.793 + 1.564)\text{mA} = 4.357\text{mA}$$

[KCL Proved]

Experiment No. 2

Introduction to Series-Parallel Circuits

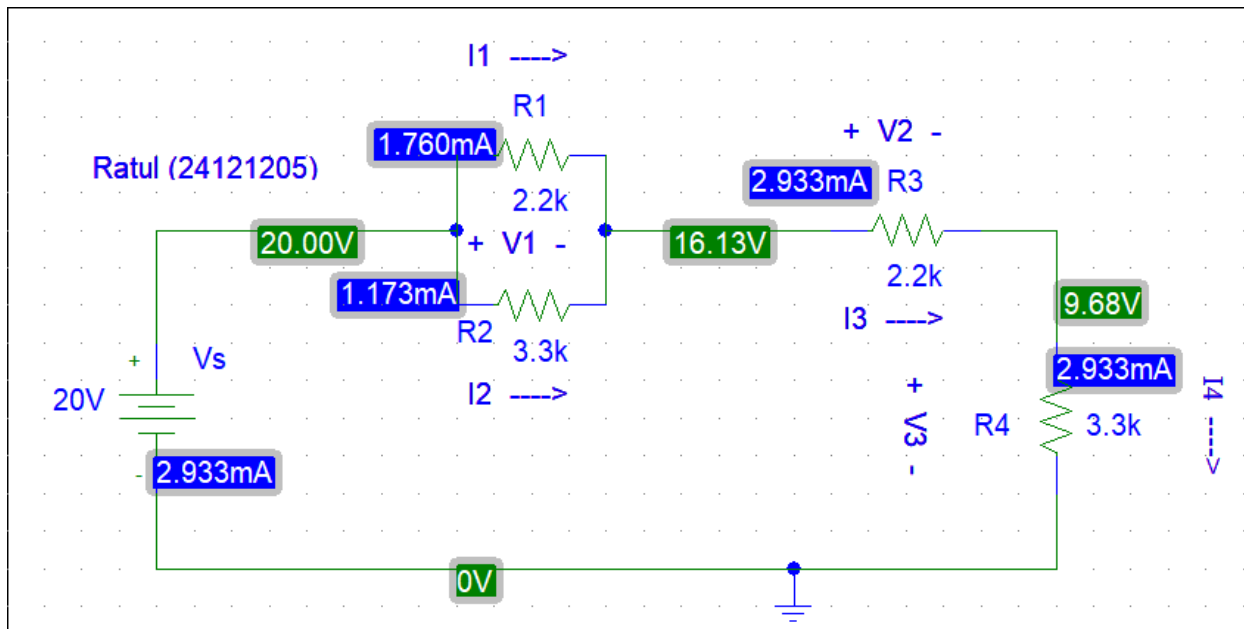


Fig-1

Here,

$$I_1 = 1.760\text{mA}$$

$$I_2 = 1.173\text{mA} \quad \text{[Parallel]}$$

$$I_3 = I_4 = 2.933\text{mA} \quad \text{[Series]}$$

And,

$$V_1 = (20.00 - 16.13)\text{V} = 3.87\text{V}$$

$$V_2 = (16.13 - 9.68)\text{V} = 6.45\text{V}$$

$$V_3 = (9.68 - 0)\text{V} = 9.68\text{V}$$

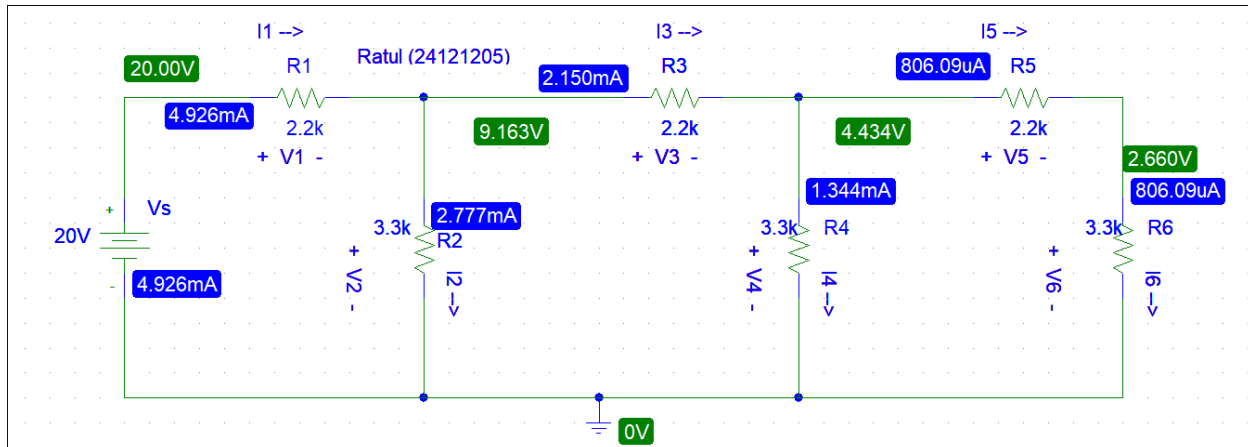


Fig-2

Here,

$$I1=4.926\text{mA}$$

$$I2=2.777\text{mA}$$

$$I3=2.150\text{mA}$$

$$I4=1.344\text{mA}$$

$$I5=I6=0.806\text{mA}$$

And,

$$V1= (20.00-9.163)=10.837\text{V}$$

$$V2=(9.163-0)=9.163\text{V}$$

$$V3=(9.163-4.434)=4.729\text{V}$$

$$V4=(4.434-0)=4.434\text{V}$$

$$V5=(4.434-2.660)=1.774\text{V}$$

$$V6=(2.660-0)=2.660\text{V}$$

Experiment No. 3

Verification of Superposition Principle

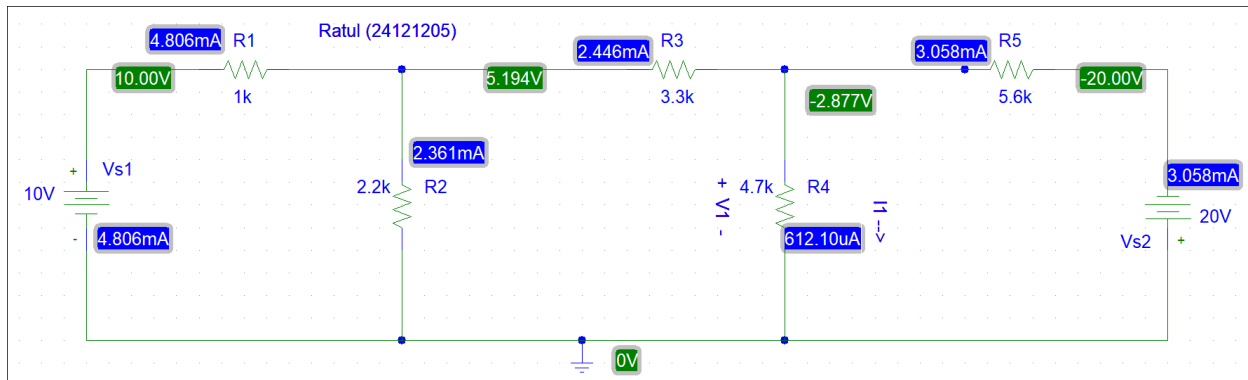


Fig-1

$$V_1 = (-2.877 - 0) = -2.877V$$

$$I_1 = -0.612mA \text{ [Opposite direction]}$$

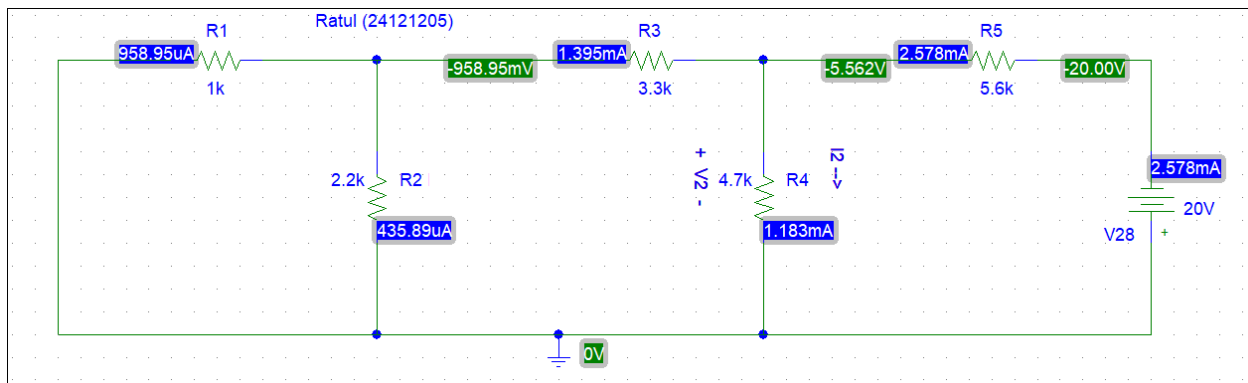


Fig-2

$$V_2 = (-5.562 - 0) = -5.562V$$

$$I_2 = -1.183mA \text{ [Opposite direction]}$$

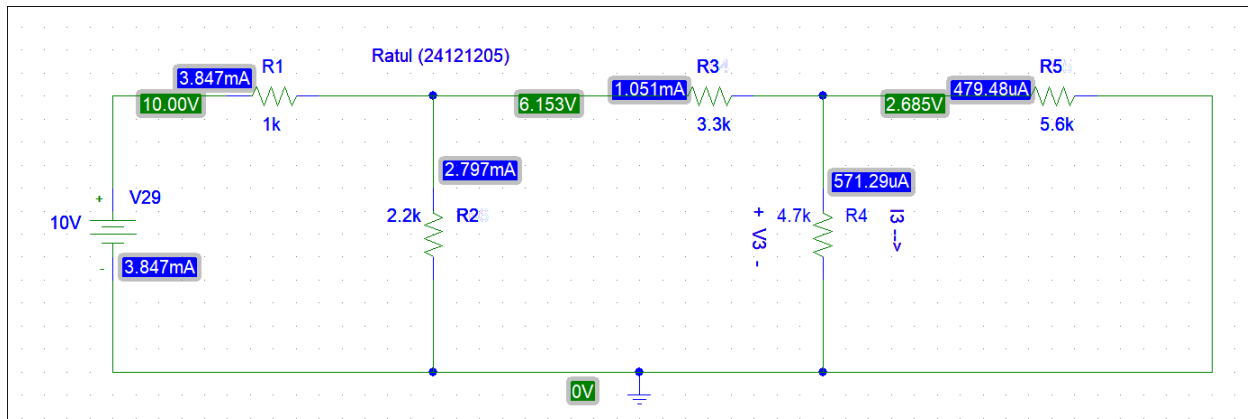


Fig-3

$$V3 = (2.685 - 0) = 2.685V$$

$$I3 = 0.571mA$$

Verification

$$V1 = V2 + V3$$

$$= (-5.567 + 2.685)V$$

$$= -2.882V \cong V1$$

$$I1 = I2 + I3$$

$$= (-1.183 + 0.571)mA$$

$$= -0.612mA = I1$$

[Proved]

Experiment No. 5

Verification of Thevenin's Theorem and Maximum Power Transfer Theorem

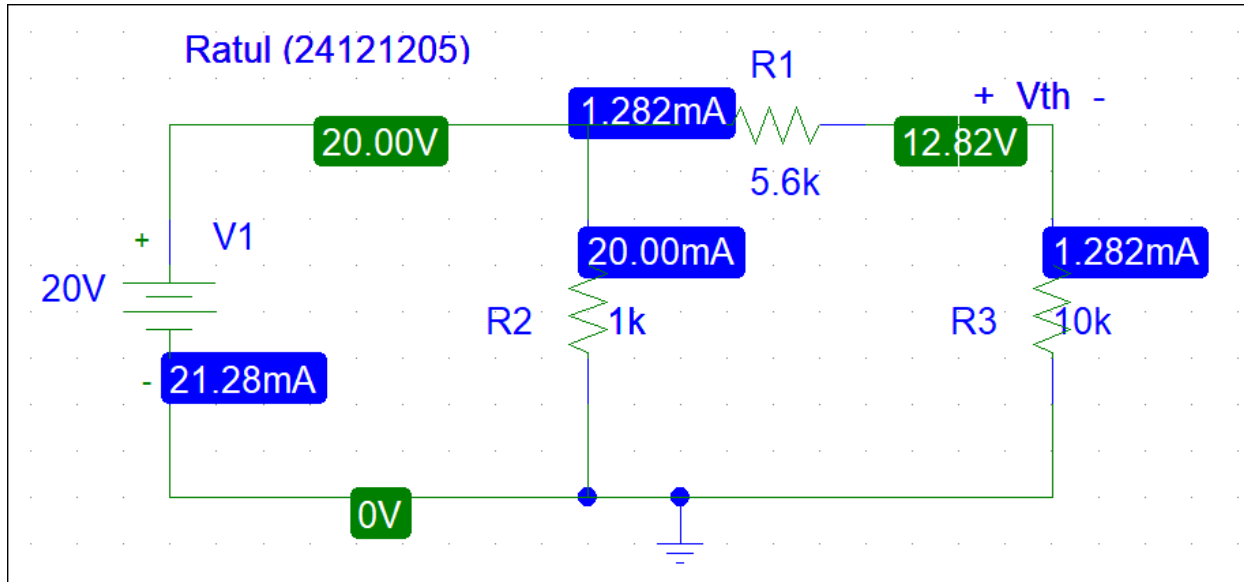


Fig-1

$V_{th}=12.82V$

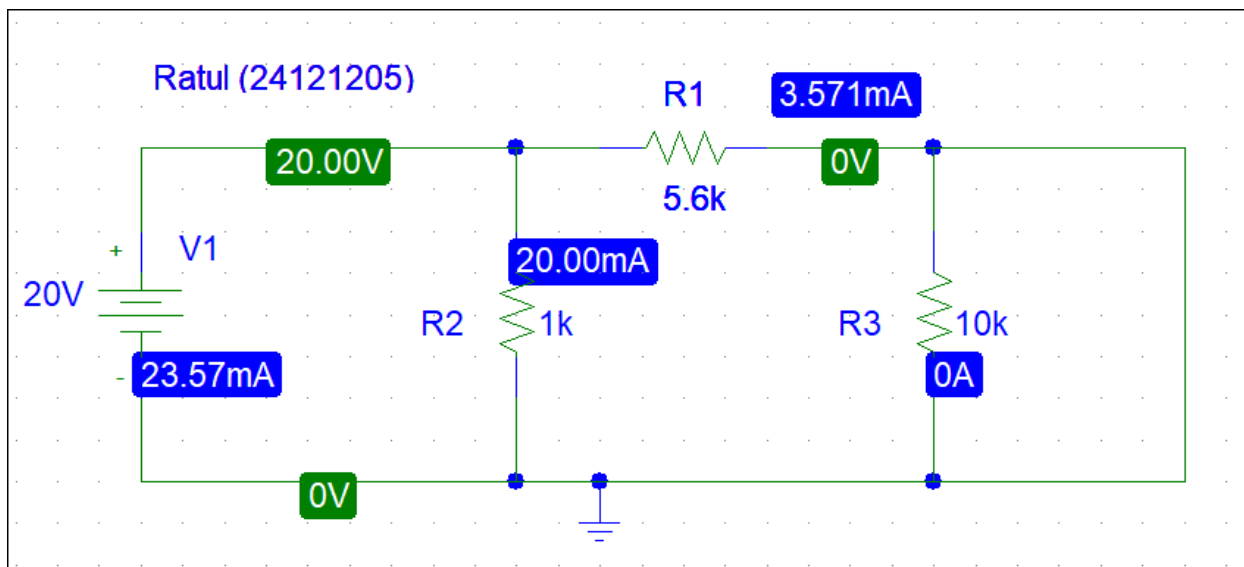


Fig-2

$I_{sc}=3.571mA$

Now, $R_{th}=V_{th}/I_{sc}=12.82/3.571=3.59k\Omega$

